

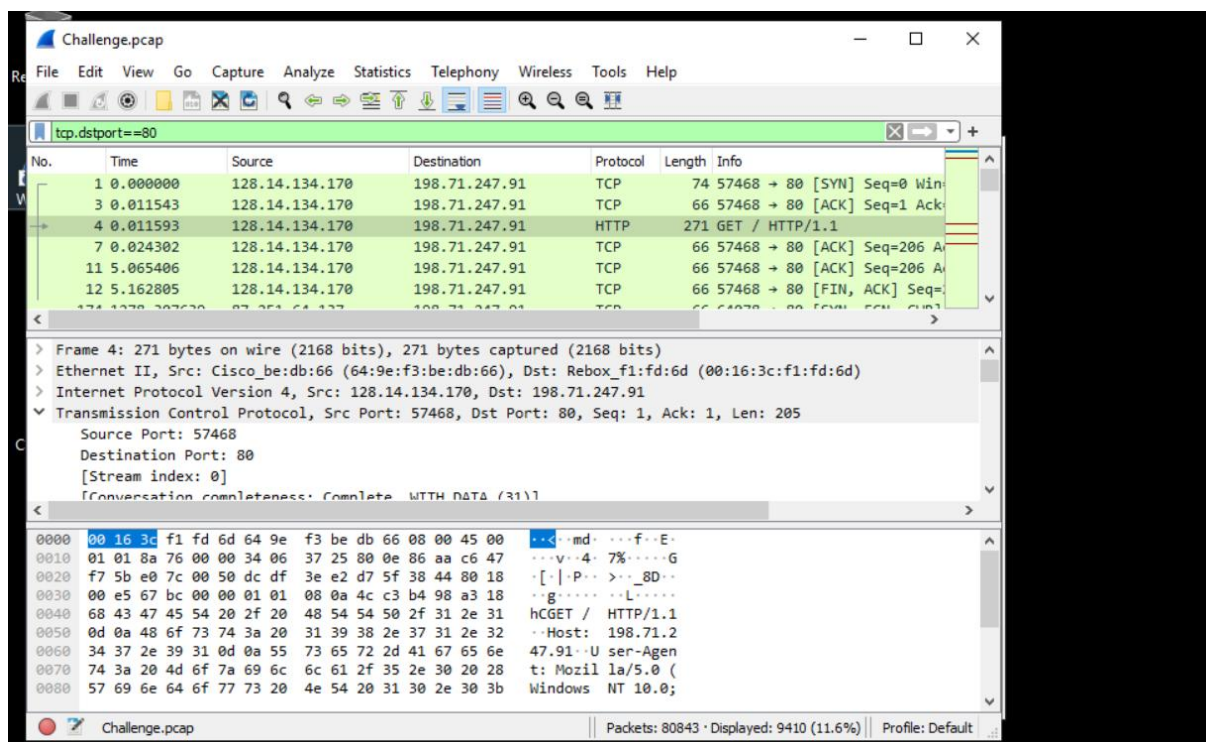
Indicators of Compromise (IOC) Report

Executive Summary

The analysis of the Challenge.pcap file reveals that the public-facing server at 198.71.247.91 was the target of reconnaissance and a sophisticated data exfiltration attack. Key findings include evidence of network scanning and, most significantly, the use of DNS tunneling to exfiltrate data, a technique often used by advanced persistent threats (APTs) after an initial compromise (such as one initiated by a Log4j exploit).

Key Asset Identified

- **Primary Target:** The server at IP address 198.71.247.91. This host is observed receiving HTTP, DNS, NTP, and other TCP connection attempts, indicating it is a multi-service, public-facing server.

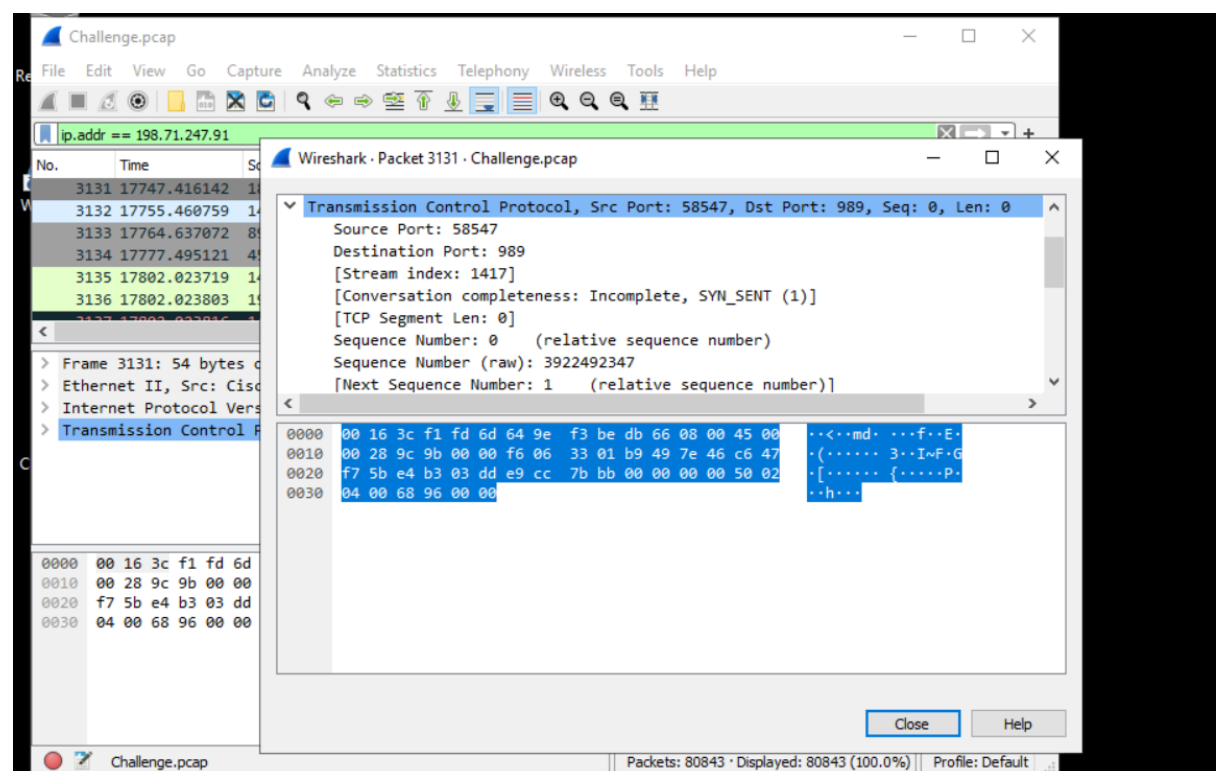


Observed Malicious Activity & IOCs

1. Network Reconnaissance / Scanning

Evidence suggests the target server was being scanned for open ports and services.

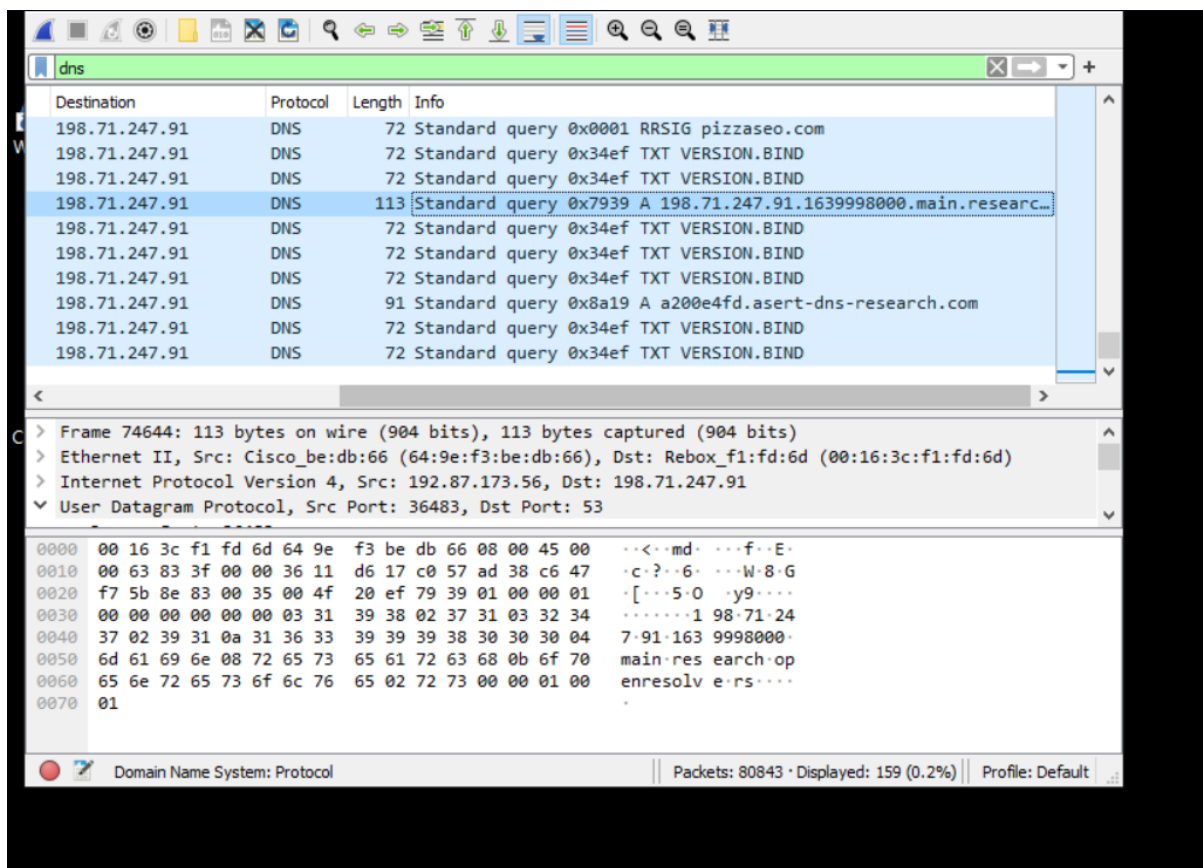
- Observation: A lone TCP SYN packet was sent to the target server on port 989 (FTPS). An incomplete three-way handshake like this is a common artifact of a port scan.
- IOC: Suspicious connection attempt to 198.71.247.91:989.



2. Data Exfiltration via DNS Tunneling

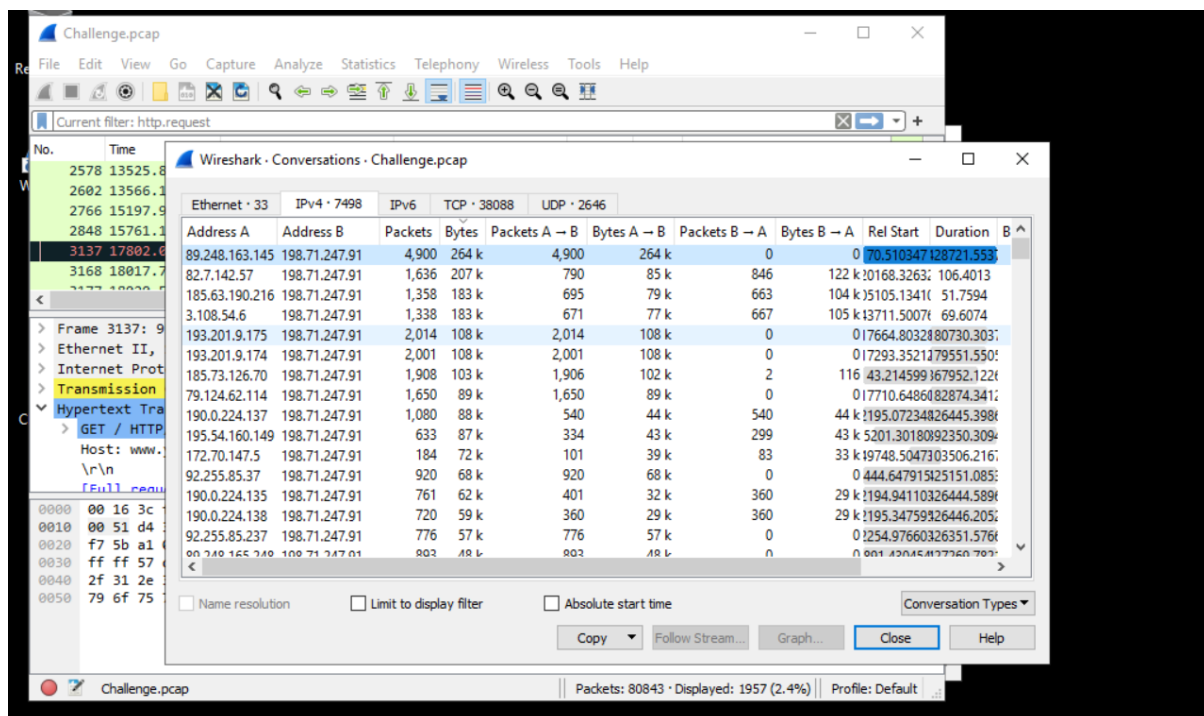
This is the most critical finding, indicating a successful compromise. Attackers encoded data into DNS queries to bypass firewalls.

- Observation: A DNS query was made for the domain 198.71.247.91.1639998000.main.research.
- Analysis: This is not a legitimate domain. It is constructed from the victim's IP address (198.71.247.91) and an epoch timestamp (1639998000), a classic signature of DNS tunneling used to exfiltrate data.
- IOCs:
 - Domain: 198.71.247.91.1639998000.main.research
 - Domain: a200e4fd.asert-dns-research.com (another suspicious query from the same capture)
 - Source IP: 192.87.173.56 (Source of the malicious DNS query)



3. Anomalous High-Volume Communication

- Observation: The Wireshark conversations window shows that the IP address 89.248.163.145 sent an unusually high number of packets (4,900) and bytes (264k) to the target server.
- Analysis: While the payload is not visible, this high volume is anomalous and warrants investigation. It could be part of a DDoS attack, data upload, or aggressive scanning.
- IOC:
 - IP Address: 89.248.163.145 (Suspicious high-volume traffic source)



Recommendations

1. Isolate: Immediately investigate the server at 198.71.247.91 for signs of compromise. It should be considered compromised until proven otherwise.
2. Block: Block the malicious domains and IP addresses identified at the network firewall.
3. Investigate: Analyze DNS logs for other queries matching this pattern to determine the scope of the data exfiltration.
4. Patch: Given the 2021 timestamp, the initial attack vector was likely the Log4j vulnerability. Ensure all Java applications on the server are patched and updated.